

Cognitively Inspired AI Memory Management System

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Abstract

Human memory is distinguished by four properties that no single existing memory system for large language model (LLM) agents fully captures: it encodes experiences in *heterogeneous formats* suited to their structure; it *organizes* those encodings adaptively across a hierarchy; it *retrieves* memories through active, iterative reconstruction rather than passive lookup; and it *utilizes* retrieved content through a compact working memory buffer rather than by dumping raw text into the prompt. We present C-AIMMS (Cognitively-inspired AI Memory Management System), a unified agentic memory framework that instantiates all four properties. C-AIMMS encodes each episodic unit in three complementary formats: a *hypergraph* capturing high-order multi-entity associations, a *visual canvas* rendering structured rich-text with adaptive information density and a *vector store* for semantic similarity. A learned adaptive selector chooses the most suitable format per episode, with Beta-Mixture-Model gating for robust fusion decisions; episodic boundaries are detected via surprise-based segmentation. Retrieval is a three-stage closed loop: a router alternates between retrieve, reflect and answer actions guided by a global evidence-gap tracker; past search trajectories are distilled into a rubric-scored experience bank that steers future retrievals; and active memory reconstruction traverses a Cue-Tag-Content hypergraph, pruning paths based on accumulated evidence. Finally, a compact online state of associative memory (δ -mem) maintains a fixed-size matrix updated by delta-rule learning, providing low-rank corrections to the frozen LLM backbone’s attention computation and keeping the prompt context uncorrupted.

1 Framework

C-AIMMS contains (1) HETREP encoding an episode into HG/VC/VS; (2) ADASTORE selecting a primary format and placing the episode in the three-tier hierarchy; (3) ITERRET showing the closed-loop router, experience bank and CTC traversal paths; (4) WORKMEM with the OSAM matrix feeding low-rank corrections into the frozen Qwen3 backbone. The system operates in two phases. During the *write phase*, each new episodic memory unit is segmented from the raw dialogue, encoded in three formats by HETREP and stored with an adaptively chosen primary format by ADASTORE. During the *read phase*, a user query triggers ITERRET, which iteratively retrieves and reflects; the retrieved signals are then fed to WORKMEM, which updates the online associative state and steers the frozen Qwen3 backbone to generate the final response.

1.1 Formalization

Let $H_T = \{(u_1, a_1), \dots, (u_T, a_T)\}$ denote a long-horizon dialogue history up to turn T , where u_i and a_i are the user utterance and agent response at turn i . Given a new user query q_T , the agent must generate a response y_T that is consistent with the accumulated history.

Episodic units. We partition H_T into a sequence of episodic units $\mathcal{U} = \{e_1, \dots, e_N\}$, where each $e_j = \{(u_{j,1}, a_{j,1}), \dots, (u_{j,m_j}, a_{j,m_j})\}$ groups a set of temporally contiguous, semantically related turns. Boundaries between episodes are detected online by ADASTORE (Section 1.3).

43 **Heterogeneous memory state.** Each episodic unit e_j is associated with up to three encod-
 44 ings: a hypergraph node set $\mathcal{H}_j$, a visual canvas image $\mathbf{I}_j \in \mathbb{R}^{H \times W \times 3}$ and a dense vector
 45 embedding $\mathbf{v}_j \in \mathbb{R}^d$. The full memory state at turn T is

$$\mathcal{M}_T = \{(\mathcal{H}_j, \mathbf{I}_j, \mathbf{v}_j, f_j)\}_{j=1}^N, \quad (1)$$

46 where $f_j \in \{\text{HG}, \text{VC}, \text{VS}\}$ is the primary storage format assigned by ADASTORE.

47 **Response generation.** Given query q_T , ITERRET produces a retrieved evidence set $\mathcal{E}_T$ via
 48 iterative retrieval. WORKMEM then updates an online associative state $\mathbf{S}_T$ using signals
 49 from $\mathcal{E}_T$ and generates the response:

$$y_T \sim p_\theta(y \mid q_T, \mathbf{S}_T), \quad (2)$$

50 where θ denotes the frozen Qwen3 backbone parameters and $\mathbf{S}_T$ steers generation through
 51 low-rank attention corrections (Section 1.5).

52 1.2 HETREP: Heterogeneous Representation

53 Human long-term memory encodes experiences in qualitatively different formats depending
 54 on their structure. Similarly, C-AIMMS maintains three complementary encodings of each
 55 episode e_j .

56 1.2.1 Hypergraph Encoding

57 We represent each episodic unit as a subgraph within a global hypergraph $\mathcal{G} = (V, \mathcal{E}_H)$ (Yue
 58 et al., 2025). The node set $V = V_T \cup V_E \cup V_F$ consists of three levels:

- 59 • *Topic nodes* V_T : thematic clusters spanning multiple episodes (ex: *career, health*).
- 60 • *Episode nodes* V_E : individual episodic units e_j , each summarized by a concise natural-
 61 language description.
- 62 • *Fact nodes* V_F : fine-grained named entities, events and assertions extracted from episode
 63 content.

64 Hyperedges $\mathcal{E}_H$ connect sets of nodes of any size: a *topic hyperedge* $h_t \in \mathcal{E}_H$ groups all
 65 episode nodes sharing the same thematic cluster; an *episode hyperedge* $h_e \in \mathcal{E}_H$ groups all fact
 66 nodes extracted from a single episode. Overlapping membership is permitted: a fact node
 67 may belong to multiple episode hyperedges and an episode node may belong to multiple
 68 topic hyperedges.

69 **Lexical-semantic indexing.** Each node $v \in V$ is associated with a BM25 lexical index over
 70 its textual content and a dense embedding $\phi(v) \in \mathbb{R}^d$ obtained from a frozen sentence en-
 71 coder. Hypergraph embedding propagation is applied: the embedding of each episode node
 72 is updated by averaging the embeddings of its hyperedge neighbors, so that semantically
 73 related but temporally distant episodes develop similar representations:

$$\phi'(e_j) = \frac{1}{|N(e_j)|} \sum_{e_k \in N(e_j)} \phi(e_k), \quad (3)$$

74 where $N(e_j)$ denotes the set of episode nodes sharing at least one hyperedge with e_j .

75 1.2.2 Visual Canvas Encoding

76 For each episode e_j , an LLM drafts a structured rich-text memory document $\text{RT}(e_j)$ in which:
 77 *crucial evidence* (e.g., entity names, dates, causal relationships) is typeset with high-visibility
 78 formatting (headings, bold, large font); *auxiliary details* (e.g., conversational filler, repeated
 79 context) are compressed into small, low-priority text or omitted. A lightweight renderer
 80 then compiles $\text{RT}(e_j)$ into a canvas image $\mathbf{I}_j$ at a chosen resolution (Shi et al., 2026).

81 This visual encoding has two key advantages. First, *adaptive information density*: the 2D layout
 82 allows crucial content to occupy more perceptual space without consuming proportionally
 83 more tokens, since a vision-language model reads the image. Second, *controllable budget*: the
 84 global information density of I_j can be adjusted by downsampling the image resolution,
 85 providing a smooth budget-fidelity tradeoff without rewriting the memory content.

86 1.2.3 Vector Store Encoding

87 We embed the summary of each episode e_j into a dense vector $\mathbf{v}_j = \text{Enc}(\text{summary}(e_j))$
 88 using a frozen sentence encoder (all-MiniLM-L6-v2). The vector store enables fast approxi-
 89 mate nearest-neighbor retrieval and serves as the primary format for episodes with weak
 90 relational or temporal structure. Hybrid retrieval combining BM25 and dense vectors via
 91 reciprocal rank fusion is used for all vector-store queries.

92 1.3 ADASTORE: Adaptive Memory Storage

93 1.3.1 Surprise-Based Episodic Segmentation

94 We detect natural episode boundaries online as dialogue tokens are processed, following the
 95 event-cognition hypothesis that surprise signals event transitions in human memory (Foun-
 96 tas et al., 2025). At each agent response token x_t , we compute the Bayesian surprise

$$\mathcal{S}_t = -\log P(x_t \mid x_1, \dots, x_{t-1}; \theta), \quad (4)$$

97 using the backbone’s own log-probabilities. A token x_t is flagged as a potential boundary if
 98 $\mathcal{S}_t > \mu_{t-\tau:t} + \gamma\sigma_{t-\tau:t}$, where μ and σ are the rolling mean and standard deviation of surprise
 99 over the past τ tokens and γ is a sensitivity hyperparameter. Graph-theoretic refinement
 100 then adjusts boundary positions to maximize within-episode cohesion using attention-
 101 key similarity as a weighted adjacency matrix. The resulting boundary set partitions the
 102 dialogue into episodes $e_1, \dots, e_N$.

103 1.3.2 Three-Tier Memory Hierarchy

104 C-AIMMS organizes memory into three tiers with different granularities and lifetimes:

105 **Short-Term Interaction Memory (STIM).** STIM buffers the four most recent dialogue
 106 turns in their raw form, enabling recency-aware fast access without retrieval overhead.
 107 When STIM reaches capacity, the oldest turn is evicted to MTEM via Least-Recently-Used
 108 (LRU) policy.

109 **Mid-Term Episodic Memory (MTEM).** MTEM stores the structured episodic units e_j ,
 110 each maintained in one or more of the three representations (HG, VC, VS). A lightweight
 111 episodic utility score

$$U(e_j) = w_1 c(e_j) + w_2 \ell(e_j) + w_3 d(e_j) \quad (5)$$

112 combines access frequency $c(e_j)$, interaction intensity $\ell(e_j)$ and recency $d(e_j)$. High-utility
 113 episodes are candidates for consolidation into LTSM.

114 **Long-Term Semantic Memory (LTSM).** LTSM stores consolidated semantic facts (user
 115 profile attributes, stable preferences, persistent knowledge) as dense embeddings in the
 116 vector store. An eligibility criterion $U(e_j) \geq \tau_u \wedge r(e_j) \geq \tau_r$ controls which episodes are
 117 promoted, keeping LTSM compact and high-precision.

118 1.3.3 Format-Adaptive Selector

119 A core design choice in C-AIMMS is that the storage format is *not* fixed uniformly across
 120 episodes. Instead, a lightweight selector $f_\theta : \mathbb{R}^d \rightarrow \{\text{HG}, \text{VC}, \text{VS}\}$ maps the feature vector
 121 $\mathbf{x}_j$ of episode e_j to its optimal primary representation.

Interaction-level features. The feature vector $\mathbf{x}_j$ encodes structural properties of episode e_j : the number of distinct named entities and their co-occurrence density (signaling relational complexity), the degree of temporal ordering in the content (signaling linear structure), the topic diversity within the episode (signaling hierarchical or visual layout needs) and the total token length. These are the same classes of features used by FluxMem (Lu et al., 2026), extended with two additional dimensions for hyperedge density and visual salience score.

Offline supervision. Since ground-truth format labels are not available, we derive supervision offline. For each episode in a held-out set of training interactions, we evaluate the downstream response quality and memory utilization effectiveness of all three formats, obtaining per-format rewards $r_j^{\text{HG}}, r_j^{\text{VC}}, r_j^{\text{VS}}$. The format with the highest reward is designated the supervision target $f_j^* = \arg \max_f r_j^f$. The selector is a two-layer MLP trained by minimizing cross-entropy:

$$\mathcal{L}_{\text{sel}}(\theta) = \sum_j \ell(f_\theta(\mathbf{x}_j), f_j^*). \quad (6)$$

BMM-gated memory fusion. When a new episode arrives, we must decide whether it should be merged into an existing MTEM unit or stored as a new one. Fixed similarity thresholds are fragile under distribution shift; instead, we use a two-component Beta Mixture Model (BMM) gate (Lu et al., 2026). Normalized matching scores $x_i \in (0, 1)$ between the incoming episode and all candidate existing units are modeled as:

$$p(x) = \pi \text{Beta}(x; \alpha_1, \beta_1) + (1 - \pi) \text{Beta}(x; \alpha_0, \beta_0), \quad (7)$$

where the two components correspond to high- and low-compatibility regimes. Parameters are estimated online via EM with responsibility-weighted moment updates. Candidates whose posterior probability under the high-compatibility component exceeds threshold τ are retained as merge targets; the incoming episode is merged into the best-scoring retained candidate or stored as a new unit if none qualify.

1.4 ITERRET: Iterative Retrieval

Standard one-shot top- k retrieval fails in long-horizon settings because: (i) a single query may not surface all necessary evidence; (ii) intermediate evidence can reveal new retrieval cues; (iii) past retrieval failures are not learned from. C-AIMMS addresses these three limitations with three interlocking mechanisms.

1.4.1 Closed-Loop Retrieval Controller

Inspired by MemR³ (Du et al., 2026), we implement a LangGraph-based closed-loop controller with three nodes:

- **Retrieve:** Queries the memory store (hypergraph, visual canvas, or vector store) with the current refined query, appending results to the evidence set $\mathcal{E}$.
- **Reflect:** Reasons over $\mathcal{E}$ against the original query q_T to identify unfilled information gaps $\mathcal{G}$ and generate a refined query for the next retrieval step.
- **Answer:** Generates the final response y_T when $\mathcal{G} = \emptyset$ or a maximum iteration budget is reached.

A *router* LLM selects the next action at each step. The global evidence-gap state $(\mathcal{E}, \mathcal{G})$ is maintained across all nodes, making the retrieval process transparent and steerable.

1.4.2 Rubric-Guided Experience Bank

Even an iterative closed-loop controller may repeat the same retrieval mistakes across queries. R²-Mem (Wang et al., 2026) observes that individual steps within historical search trajectories can be high- or low-quality independently of the trajectory’s final outcome and proposes scoring them with a fine-grained rubric evaluator.

We adopt this idea by maintaining an offline *experience bank* $\mathcal{B}$. After each retrieval episode completes, a rubric evaluator scores each Retrieve and Reflect step along dimensions including query specificity, evidence coverage and gap-gap redundancy. High-quality steps and low-quality steps (along with the rubric rationale for each) are distilled into the bank via a self-reflection learner that abstracts step-conditional guidance. During inference, the top- k most similar experiences from $\mathcal{B}$ are retrieved by query similarity and injected into the router’s context, biasing action selection toward known high-quality behaviors and away from known failure modes.

1.4.3 Active Hypergraph Reconstruction

The closed-loop controller issues queries to the memory store, but the store itself may be queried passively (top- k nearest neighbors) or actively (path-following with pruning). We implement active retrieval on the hypergraph using the Cue-Tag-Content (CTC) traversal of MRAgent (Ji et al., 2026).

Each node in $\mathcal{G}$ is augmented with *associative tags*: short descriptors that semantically connect fine-grained cues to the content of adjacent nodes. Tags are intermediate-resolution anchors: they are visible before the agent commits to loading the full content of a node, allowing the router to decide whether a traversal path is worth pursuing.

Retrieval proceeds as follows:

1. *Cue extraction*: The current evidence-gap state $(\mathcal{E}, \mathcal{G})$ is used to extract retrieval cues - named entities, time expressions, causal predicates - that index into the hypergraph.
2. *Tag-guided expansion*: For each cue, the router reads the associative tags of neighboring nodes and selects the most promising next hops.
3. *Content loading*: The content of selected nodes is loaded and appended to $\mathcal{E}$.
4. *Pruning*: Paths whose tags are inconsistent with the current gap $\mathcal{G}$ are pruned before content is loaded, avoiding combinatorial explosion.
5. *Evidence-triggered stopping*: The process halts when $\mathcal{G}$ reaches sufficiency or the iteration budget is exhausted and control returns to the closed-loop controller.

Routing across formats. For episodes stored in VC format, retrieval is performed by submitting the visual query (rendered from the current evidence-gap state) to a multimodal reader that scans the canvas image. For VS-format episodes, standard dense retrieval is used. The closed-loop router decides which format to query at each step based on the nature of the current gap.

1.5 WORKMEM: Working Memory Utilization

Once the closed-loop controller emits Answer with evidence set $\mathcal{E}_T$, we must translate $\mathcal{E}_T$ into influence on the Qwen3 backbone. The naive approach—injecting $\mathcal{E}_T$ as additional prompt tokens—consumes token budget, suffers from retrieval noise entering the context and causes context rot when multiple retrieved segments interact. Instead, C-AIMMS routes memory influence through δ -mem (Lei et al., 2025).

Online state of associative memory (OSAM). δ -mem maintains an online state $\mathbf{S}_t \in \mathbb{R}^{r \times r}$ per Transformer layer, where $r \ll d$. The state encodes key-value associations from the retrieved evidence. For each retrieved fact $m \in \mathcal{E}_T$, memory projections map the hidden state to low-dimensional memory keys and values:

$$\mathbf{k}_t^m = L_2\text{norm}(\tanh(\mathbf{W}_k^m \mathbf{x}_t)), \quad \mathbf{v}_t^m = \mathbf{W}_v^m \mathbf{x}_t. \quad (8)$$

The state is updated by gated delta-rule learning:

$$\mathbf{S}_t = \lambda_t \mathbf{S}_{t-1} + \beta_t (\mathbf{v}_t^m - \mathbf{S}_{t-1} \mathbf{k}_t^m) (\mathbf{k}_t^m)^\top, \quad (9)$$

where $\lambda_t = 1 - \beta_t$ is a data-dependent forget gate. This update writes only the residual along the key direction: associations already encoded induce small updates, while new associations are written strongly.

Algorithm 1 C-AIMMS Inference Pipeline**Require:** Query q_T , history H_T , memory state $\mathcal{M}_{T-1}$, experience bank $\mathcal{B}$, OSAM $\mathbf{S}$

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1: Write phase (per new turn):
2: Detect episode boundaries via surprise-based segmentation (§1.3.1)
3: for each new episode  $e_j$  do
4:   Encode  $e_j$  into  $(\mathcal{H}_j, \mathbf{I}_j, \mathbf{v}_j)$  via HETREP (§1.2)
5:   Select primary format  $f_j^* \leftarrow f_\theta(\mathbf{x}_j)$  (§1.3.3)
6:   Store  $e_j$  via BMM-gated fusion into MTEM (§1.3)
7:   Promote high-utility episodes to LTSM (§1.3.2)
8: end for
9: Read phase (per query):
10: Initialize  $\mathcal{E} \leftarrow \emptyset, \mathcal{G} \leftarrow \{q_T\}$ 
11: Retrieve top- $k$  experiences from  $\mathcal{B}$  by similarity to  $q_T$ 
12: repeat
13:   action  $\leftarrow$  ROUTER( $q_T, \mathcal{E}, \mathcal{G}$ , experiences)
14:   if action = Retrieve then
15:     Retrieve via CTC-traversal (§1.4.3) or dense search; update  $\mathcal{E}$ 
16:   else if action = Reflect then
17:     Reflect on  $(\mathcal{E}, \mathcal{G})$ ; refine  $\mathcal{G}$  (§1.4.1)
18:   end if
19: until action = Answer or budget exceeded
20: Populate OSAM  $\mathbf{S}$  from  $\mathcal{E}_T$  via delta-rule (§1.5)
21: Generate  $y_T \sim p_\theta(y \mid q_T, \mathbf{S})$  with low-rank attention corrections
22: Update experience bank  $\mathcal{B}$  with rubric-scored trajectory (§1.4.2)
23: return  $y_T$ 

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211 **Low-rank attention steering.** At generation time, the online state produces a read vector
 212 $\mathbf{r}_t = \mathbf{S}_{t-1} \mathbf{q}_t^m$ by querying with the current token’s memory query. This read vector is
 213 projected into two corrections applied to the frozen backbone:

$$\Delta \mathbf{Q}_t = \mathbf{W}_Q^\Delta \mathbf{r}_t, \quad \Delta \mathbf{O}_t = \mathbf{W}_O^\Delta \mathbf{r}_t. \quad (10)$$

214 The query correction $\Delta \mathbf{Q}_t$ biases the attention pattern toward memory-relevant keys in
 215 the current context; the output correction $\Delta \mathbf{O}_t$ directly augments the attention output.
 216 Crucially, the backbone parameters θ are never modified and the prompt context q_T carries
 217 no retrieved content.

218 **Interaction with ITERRET.** The OSAM is populated in two phases: first, from the retrieved
 219 evidence set $\mathcal{E}_T$ (offline, before generation begins); second, incrementally from the model’s
 220 own newly generated tokens during response generation (online). This allows C-AIMMS
 221 to leverage both external retrieved memory and the model’s in-context reasoning within a
 222 single unified state.

223 2 Experimental Setup

224 **Benchmark.** We evaluate C-AIMMS on **LoCoMo** (Maharana et al., 2024), a long-horizon
 225 conversational memory benchmark comprising very long dialogue traces (average ~ 300
 226 turns, $\sim 9,000$ tokens, spanning 19-35 sessions). Questions are grouped into five categories:
 227 *single-hop* (answer in one session), *multi-hop* (answer requires aggregating across sessions),
 228 *temporal* (reasoning over time gaps), *open-domain* (combining dialogue facts with world
 229 knowledge) and *adversarial* (recognizing unanswerable questions).

230 **Backbone.** We use **Qwen3-4B-Instruct** (Qwen Team, 2025) as the frozen backbone for
 231 all modules that require LLM calls (router, reflector, selector supervision, rubric evaluator,
 232 visual canvas drafting and response generation).

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